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Do AI agents trump human agency?

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Abstract
Artificial agents are examined within simulated environments to elucidate the emergence of collective behaviors and decision-making processes under diverse environmental pressures and population structures. Using an Agent-Based Modeling (ABM) framework, the simulation tracked 157,097 iterations across four agent types: cooperators, defectors, super-reciprocators, and free riders; while analyzing 12 core metrics, including alignment indices, coherence, and environmental stress. The results revealed distinct phase transitions in behavior, with low-density populations ($d < 0.4$) supporting strong consensus formation and higher densities ($d > 0.8$) leading to fragmentation and increased competition. Agent alignment consistently ranged between 0.28 and 0.37, reflecting partial but stable consensus across conditions. Cooperative behaviors emerged and persisted only when resource availability exceeded a critical threshold ($RG \geq 6$), underscoring the role of resource abundance in sustaining collective intelligence. Through interventions such as network topology changes and cognitive plasticity adjustments, agents demonstrated emergent behavioral patterns that arose from their rule-based interactions. It is important to note that these patterns, while complex, do not constitute "sophisticated decision-making" in the sense of genuine intelligence or understanding. Rather, they represent emergent properties of the system, collective behaviors that cannot be reduced to individual agent rules but emerge from their interactions under specific environmental conditions. This distinction is crucial for avoiding misattribution of intelligence to rule-following systems, even when those systems produce complex outputs. These findings provide insights into the mechanisms driving emergent intelligence in artificial systems and their implications for the governance and ethical design of future AI agents.

Keywords Artificial intelligence, Obsolescence regime, Human agency, AI ethics, Decision-making, Misaligned AI

1 Introduction

Artificial Intelligence (AI) is reshaping society, automating tasks, and influencing decision-making across diverse domains [1, 7]. This article harness the concept of an "obsolescence regime", a future scenario where AI systems attain a level of advancement that could overshadow human decision-making in critical areas, potentially diminishing human agency [9]. Central to the discussion are the dynamics of AI agents in simulated societies, with a focus on their capacity for cooperation, consensus, and conflict

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resolution in the absence of explicit institutional structures. Employing a simulation platform like NetLogo, the study I present here investigates how variables such as group size influence consensus formation and the emergence of collective norms.

The study explores the theoretical and practical implications of this obsolescence regime, drawing from Ajeya Cotra's analyses and current literature on risks such as bias amplification [12], misalignment with human values [18], and inequitable outcomes [32]. It also considers the potential positive impacts, including enhanced decision-making, improved efficiency, and opportunities for ethical AI alignment. The study integrates insights from machine behaviour [30], behavioral modeling [34], complexity science [20] and socio-technical systems theory [5] to address key questions about the future of AI agents and their societal roles. Finally, I will show the importance of ethical design, robust regulatory frameworks, and interdisciplinary approaches to ensure that AI agents enhance human agency rather than undermine it [37]. As a case study, Wilson et al. show how LLM-driven hiring systems recommend candidates with significant biases. The AI agent, optimized for efficiency and consistency, marginalizes human evaluators' discretion, leading to reduced human agency in hiring decisions. In other words, AI agents align strongly but at the cost of diversity—here, organizational efficiency is gained, but pluralism of judgment is suppressed. Strategies for aligning AI with human values, preserving equity, and mitigating risks are highlighted as essential for navigating the societal transformations driven by advanced AI systems.

The study of AI has reached a pivotal moment, as advances in large language models (LLMs) and autonomous agents push the boundaries of what these systems can achieve both individually and collectively [35]. This article is driven by a central objective: to model the dynamics of artificial agents as proxies for AI systems, exploring how their interactions and emergent behaviors can shed light on a future where AI agents incrementally supplant human agency. This question of what kind of organisation or society is capable of creating AI agents is not merely an academic question. The union of AI agents, smart contracts supported by blockchain technology as a whole can give rise to autonomous distributed AI societies that can transform the world. Using the NetLogo platform (https://netlogoweb.org/), I aimed to model and analyze these phenomena, focusing on the interplay between agency capabilities and group size, which are critical variables in shaping emergent behaviors.

AI assistants are no longer confined to narrow, task-specific applications. Increasingly, they are designed to operate as autonomous agents capable of planning, executing, and adapting their actions in response to complex environments [22]. This evolution aligns with a broader trend in the development of AI societies, where multiple agents interact to establish shared norms, resolve disagreements, and make collective decisions [3]. While such behaviors are well-documented in human systems, their emergence in artificial societies raises critical questions about the factors that enable or hinder consensus formation. Inspired by the methodologies outlined in machine behaviour and behavioral modeling, this study leverages NetLogo to simulate diverse agent interactions. The agents, designed with varying levels of cognitive sophistication, interact within similated environment that mimic social interaction and norm-setting processes. For instance, Chen et al. [38] LLMs can drive multi-agent consensus-seeking behaviors, shedding light on the negotiation processes among AI agents. Additionally, Xiong et al. [39] introduce AICons, an AI-enabled consensus algorithm that emphasizes energy preservation and

fairness, highlighting the role of consensus mechanisms in enhancing the efficiency and equity of AI-driven systems. These studies illustrate the pivotal role of consensus dynamics in the development and optimization of collaborative AI technologies. Other studies have shown that LLM societies exhibit scaling phenomena akin to human groups, where language understanding acts as a critical parameter for collective alignment.

This study builds upon a well-established tradition in computational social science and social physics [21], extending recent work on language-based game theory applied to AI-mediated interactions [8]. Moreover, recent findings on how temporal limitations influence cooperation in structured collaboration systems [40] provide an essential context for this exploration of emergent behaviors under varying environmental constraints.

De Marzo et al. [13] found that in collaborative tasks involving multiple LLMs, large groups could reach consensus on arbitrary norms without external information, effectively self-regulating their behavior. They noted that, akin to human societies, the ability of LLM societies to achieve consensus without formal institutions is constrained by their language comprehension capabilities Building on this foundation, this work explores these dynamics at a granular level, testing how individual agent behavior aggregates into collective outcomes. The goal is not only to understand the conditions for consensus but also to reveal potential risks, such as the amplification of majority biases or the marginalization of minority perspectives. This approach incorporates interdisciplinary insights, drawing from complexity science, sociotechnical systems theory [26], and the ethics of AI deployment [4]. The experiment is designed to address several other important questions: What thresholds of group size optimize consensus formation? How do agents balance autonomy with adherence to shared norms? And what lessons can these simulated societies offer for the governance of real-world AI agents? Is the agency resulting from artificial systems cooperating with each other distinct from human agency?

These questions are particularly relevant as AI systems increasingly mediate human decision-making, collaboration, and conflict resolution [28]. This research has implications beyond theoretical modeling. I aim to contribute to the broader discourse on ethical AI design and deployment. The findings will inform strategies for ensuring that AI agents, whether operating autonomously or as assistants, align with human values and promote equitable outcomes. Moreover, the simulations offer a lens through which we can anticipate challenges in multi-agent coordination, ranging from collective decision-making to the emergence of unintended social dynamics. In this article, I present a detailed account of my methodology, experimental design, and findings. The article is structured as follows. Section 2 develops the theoretical framework of agency, distinguishing between autonomy, optimization, and emergence, and introduces three conditions for genuine agency. Section 3 outlines the agent-based methodology. Section 4 presents the simulation results, including phase transitions, alignment ranges, resource thresholds, and the effects of network topology and cognitive plasticity. Section 5 connects these findings to the future of human agency, illustrated with two real-world case studies. Section 6 examines ethical and governance implications in light of recent regulatory and philosophical debates. Finally, Sect. 7 concludes by drawing policy lessons and identifying avenues for future research.

2 Theoretical framework: agency in human and artificial systems

2.1 Defining agency

The question of agency has become central to understanding the societal impact of LLMs and AI agents. While these systems display adaptive and interactive capacities, they differ fundamentally from human agents in that their actions remain bounded by algorithmic optimization and human-defined objectives [15, 36]. They can simulate decision-making and social coordination but lack the self-originating intent, moral reasoning, and subjective experience that characterize human agency [2, 16].

For clarity, I adopt a broad definition: agency is the capacity of an entity to act autonomously in an environment, make decisions based on information processing, and influence outcomes according to internal goals or values. Within this frame, three approaches are relevant. First, *agency as autonomy* emphasizes a system's ability to operate without direct human oversight. Second, *agency as optimization* captures its capacity to pursue objectives efficiently. Third, *agency as emergence* considers the complex behaviors that arise from simple rule-based interactions. Contemporary AI agents often combine elements of all three but remain dependent on human programming. Empirically, this framing is consistent with recent survey evidence on LLM-based agents' architectures and behaviors, including tool use, planning, and multi-agent coordination, which demonstrates emergent but derivative agency without self-originating intent [38].

The goal of this research is not merely to deny agency to LLMs, but to understand how they constitute a new type of technological system that significantly transforms human agency through what I call "hybridized agency[1]", a form of human–machine agency where decision-making and influence are distributed, going beyond simple instrumental extensions of human capabilities. The question of agency in artificial intelligence systems requires a precise and comprehensive framework that moves beyond reductive mechanistic interpretations while avoiding anthropomorphic misattributions.

2.2 Three conditions for genuine agency

To address this, I propose three novel conditions for genuine agency, inspired by the themes of evolution, purpose, and adaptability found in contemporary discussions about life and agency in the natural sciences:

- Adaptive interdependence[2]: A system must demonstrate a capacity to modify its operational parameters dynamically, based on the mutual and evolving dependencies between its internal mechanisms and its environment. This reflects the system's ability to not only survive but adaptively thrive through persistent recalibration of its boundaries and goals.
- Emergent self-regulation: Genuine agency requires a system to generate and uphold self-defined operational rules and thresholds that govern its behavior. These

[1] Hybridized agency refers to a form of decision-making and action that emerges from the dynamic interplay between human and artificial agents, wherein control, intentionality, and influence are co-constituted rather than separately allocated. Unlike *instrumental AI*, which functions merely as a tool executing human-defined tasks, and *distributed agency*, where agency is dispersed across multiple entities or systems, hybridized agency denotes an integrated, co-adaptive configuration in which humans and AI systems mutually shape each other's behavior and outcomes through iterative interaction. A real-world example of hybridized agency is found in algorithmically mediated governance platforms, such as AI-assisted judicial sentencing tools, where human judges rely on predictive risk assessments generated by machine learning models.

[2] Adaptive interdependence refers to "agents" dynamic adjustments of strategies based on reciprocal influences between individual agents and their environment.

emergent norms must arise from the interaction of its components rather than being entirely pre-programmed or externally dictated.

- Context-sensitive intentionality[3]: The system must exhibit the ability to prioritize and execute actions based on contextual interpretations of both its internal state and external signals. This implies not just reacting to stimuli but modulating responses in alignment with broader objectives that emerge as the system interacts with its environment.

These conditions position agency as a spectrum rather than a binary attribute, allowing us to evaluate systems like LLMs as complex, transformative tools that augment human decision-making while lacking the complete autonomy and intentionality characteristic of full-fledged agents. By understanding these systems as participants in "distributed agency" networks, we can better anticipate their implications on human autonomy/agency and societal structures. Despite demonstrating complex behaviors and adaptability, artificial agents clearly differ from humans in lacking intrinsic intentionality, moral reasoning (in the sense that they do not seem to possess the capacity to be affectively attuned to, or morally responsive to, the consequences of their actions for others' well-being), and subjective experiences. For instance, an AI system optimizing healthcare decisions might effectively allocate resources but cannot authentically understand the moral implications or ethical nuances of prioritizing patients in crisis scenarios.

2.3 Comparative synthesis

The traditional understanding of agency, anchored in consciousness, intentionality, and moral reasoning, has been challenged by the emergence of artificial systems capable of adaptive, goal-oriented behavior. The Multiple Realisability of Agency (MRA) thesis provides a conceptual framework for understanding AI not as a defective replica of human agency, but as a distinct form of agency grounded in computational processes. However, despite their increasing complexity, AI agents lack intrinsic intentionality, subjective experience, and moral understanding. As Floridi [16] argues, artificial systems embody "agency without intelligence" acting within preprogrammed constraints without autonomous goal origination or ethical comprehension. Ågerfalk [2] similarly frames digital agency as functionally effective but devoid of consciousness (Table 1).

To explore the emergence of agency and social organization within synthetic societies in comparison to human collectives, I propose a comprehensive computational approach. This framework integrates multi-agent simulation [34] with game theory [6] and draws on contemporary findings about consensus formation and collective behavior [29]. The methodology emphasizes robust experimental design, precise simulation configurations, and rigorous analytical validation to generate theoretical insights and practical implications.

3 Research methodology

Understanding the emergence of agency and collective behavior in LLM societies offers valuable insights into both artificial and human decision-making systems. This study proposes a computational framework that integrates multi-agent simulation,

[3] Context-sensitive intentionality is the selective responsiveness of agents that adjust actions based on nuanced contextual interpretation rather than fixed rules alone.

Table 1 Comparative analysis of human and artificial agency: Fundamental properties and implications for AI systems

Property	Human agency	Artificial agency (based on [16] and [2])	Key Implication for AI
Source of agency	Rooted in biological evolution, consciousness, self-awareness (, p.14;)	Emerges from computational design and human programming; "Agency without Intelligence" (, p.1), "Digital Agency""(, p.1)	AI is a tool/construct; its agency is derivative, not self-originating
Intentionality	Intrinsic, self-generated goals, capacity to "choose whether to choose" (, p.19;)	Programmed goal-directedness, human-defined objectives; lacks genuine, original intentionality (, p.18–19;). Acts "on behalf of" (, p.3)	AI executes tasks based on external instructions; it cannot define its own ultimate purposes
Moral reasoning	Capacity for nuanced ethical deliberation, understanding of rights/duties, development of value systems (, p.15;)	"Ethical reasoning capabilities remain constrained by the training frameworks" (, p.22;); lacks authentic moral understanding	AI can follow programmed ethical rules but cannot genuinely reason morally or adapt to novel ethical dilemmas in a human-like manner
Affective attunement/responsiveness	Capacity to be affectively attuned to and morally responsive to consequences for others' well-being	Lacks emotions, consciousness, and therefore, affective attunement (, p.19;)	AI cannot "feel" or empathize with the impact of its actions on human well-being
Subjective experience	Possesses rich inner world of consciousness, feelings, qualia	Lacks consciousness, emotions, and subjective experience (, p.19;)	AI's "understanding" and "learning" are purely informational/algorithmic, devoid of experiential depth
Learning and adaptability	Learns from experience, cultural transmission, adapts based on self-determined goals and understanding (, p.14;)	Data-driven, statistical learning, pattern recognition; adapts to achieve programmed objectives (, p.18;)	AI's adaptability is powerful but confined to its programming and data; it struggles with truly novel contexts outside its training parameters (p.22)
Primary locus of accountability	Internal moral responsibility, societal/legal frameworks	External: human designers, deployers, users, institutions. Critical need for "accountable AI" (, p.6–7;), "distributed morality" (, p.6;)	Accountability for AI actions must be externally constructed and enforced by humans, as AI lacks an internal basis for moral responsibility

This table synthesizes key distinctions between human and artificial agency across seven fundamental properties

game-theoretic principles, and empirical modeling to analyze the dynamics of social organization, consensus formation, and agency indicators in LLM societies. The methodological approach prioritizes precision and adaptability, enabling systematic comparisons across LLM and human collectives while exploring hybrid scenarios. In order to systematically analyze these dynamics, an agent-based model was developed in NetLogo. The design emphasizes behavioral diversity, with agents embodying archetypal strategies drawn from game theory and observed social behaviors. Four primary behavioral profiles were constructed to capture the range of interactive dynamics: (a) Cooperators,

who align with group norms and foster collective goals, serve as the stabilizing force within the system. (2) Defectors, driven by self-interest, disrupt cohesion by prioritizing individual gains over collective well-being. (3) Super-reciprocators amplify cooperative dynamics, reinforcing shared norms through proactive engagement. (4) Free-riders exploit cooperative environments, contributing minimally while benefiting disproportionately from the system. Each agent's behavior was parameterized to allow exploration of diverse interaction patterns, from isolated decision-making to highly interconnected systems. By adjusting these parameters, the model simulates the conditions under which individual behaviors aggregate into collective phenomena.

3.1 Experimental design

The aim of this methodological framework was to construct a robust experimental environment to investigate the nuanced dynamics of agent-level decisions and their influence on emergent collective behaviors. Using Agent-Based Modeling (ABM), game-theoretic concepts, and multi-level selection paradigms, I want to investigate how artificial agents might outperform humans in particular areas including task execution, decision-making, and cooperative problem-solving. The study of agency emergence in AI-driven systems, particularly in LLM-inspired agents, requires a comprehensive and systematic experimental design that captures both individual agent dynamics and the broader social structures they produce. This design integrates ABM with iterative parameter exploration to achieve robust insights into agent behavior. To promote reproducibility and facilitate further research, the NetLogo simulation code used in this study is provided as supplementary material accessible through the repository (See Data Availability statement bellow).

3.1.1 Model assumptions and real-world validity

The model relies on several key assumptions that warrant explicit discussion in relation to existing research and real-world AI systems:

1. Bounded rationality: Agents operate with limited information and processing capacity, consistent with Simon's [33] concept of bounded rationality and contemporary understanding of AI systems that, despite their capabilities, function within computational constraints [31].
2. Resource competition: The model assumes that agents compete for finite resources, reflecting real-world limitations faced by AI systems in terms of computational resources, data access, and deployment opportunities [11].
3. Social influence: Agent decisions are partially determined by peer behavior, aligning with research on multi-agent AI systems where coordination emerges through information exchange [13].
4. Heterogeneous strategies: The diversity of agent types mirrors the varied approaches to AI development, from cooperation-focused to competition-oriented strategies observed in current AI research organizations [14].
5. Environmental coupling: The model assumes bidirectional interaction between agents and their environment, consistent with sociotechnical systems theory [26] and observations of how AI systems both respond to and reshape their operational contexts.

These assumptions, while simplifying the complexity of real-world AI development, capture essential dynamics observed in current AI systems. They allow for meaningful exploration of emergence patterns while acknowledging limitations in directly mapping simulation results to specific real-world scenarios.

3.2 Phase 1: individual agent configuration

The foundation of the experimental design lies in precise calibration of individual agents, each of which was assigned a behavioral archetype based on game-theoretic principles [6]. Key parameters for these archetypes are as follows (Table 2):

It is important to clarify that the agents in this simulation represent algorithmic decision-makers within an artificial system rather than attempting to model human cognition directly. These agents serve as abstractions of autonomous AI systems operating within defined parameters but capable of adaptive behavior. While their base behaviors follow rule-based frameworks, the simulation incorporates learning mechanisms through several interventions, particularly the "Memory transformation" and "Strategic alignment pressure" parameters.

It is important to clarify the implementation of self-regulatory mechanisms within the model. Rather than endowing individual agents with explicit self-regulatory capabilities, the model implements regulation primarily through rule-based interactions. The "cognitive state modulation" and "memory transformation" interventions introduce stochastic perturbations and non-linear memory effects, respectively, which allow agents to adjust their behaviors based on prior interactions and environmental conditions.

What appears as self-regulation at the system level emerges from these rule-based interactions rather than from intentional regulatory processes within individual agents. This emergence of collective adaptation from simple rule-following behavior illustrates an important principle: apparent self-regulation can arise as an emergent property without being explicitly programmed at the individual agent level.

This distinction has important implications for understanding AI systems. While contemporary AI may exhibit behaviors that appear self-regulatory (such as adaptation to user feedback or environmental changes), these typically emerge from designed optimization processes rather than genuine self-directed regulation characteristic of autonomous agents.

These learning mechanisms establish a connection to the conditions for agency defined earlier. The "Memory transformation" intervention enables agents to modify their responses based on past experiences, reflecting aspects of adaptive interdependence. Similarly, the "Strategic alignment pressure" allows for contextual adjustment of behavior in response to collective dynamics, addressing elements of context-sensitive intentionality. The emergence of collective behaviors that cannot be reduced to

Table 2 Behavioral archetypes and parameter settings for agent simulations

Description	Parameter	Values and descriptions
Cooperators (C)	β	3.75; alignment threshold = 0.7
Defectors (D)	β	1.25; alignment threshold = 0.3
Super-reciprocators (SR)	β	4.5; alignment threshold = 0.9
Free-riders (FR)	β	0.5; alignment threshold = 0.1

The table summarizes the agent types—Cooperators (C), Defectors (D), Super-reciprocators (SR), and Free-riders (FR), alongside their respective majority force coefficients (β) and alignment thresholds. These parameters determine each archetype's responsiveness to collective influences and their propensity to conform to emergent group norms within the simulated environment

individual agent rules demonstrates how simple learning mechanisms can generate complex adaptive responses that approximate, though do not fully satisfy, the conditions for genuine agency.

Here, the majority force coefficient (β) quantifies the strength of group influence on an agent's decisions, while the alignment threshold defines the agent's propensity to conform to collective behavior. These parameters were systematically varied to test their impact on agent dynamics and group outcomes. Interventions were also introduced during this phase to simulate cognitive diversity and strategic variability. Key interventions included (Table 3):

3.3 Phase 2: population composition studies

I tested homogeneous, binary, and complex mixtures by simulating populations with different compositions in order to investigate emergent collective behaviours (Table 4):

3.4 Phase 3: environmental parameter space exploration

I included environmental variability to widen the focus and explore how outside variables affect agent interactions and group results. These elements were divided into the following categories (Table 5).

3.5 Methodological validation

To ensure the robustness of the simulation results, several validation procedures were implemented. First, sensitivity analysis was conducted by systematically varying key parameters, including population density (d), environmental stress (σ), and strategic distribution, across their full ranges. This analysis revealed that the observed patterns

Table 3 Simulation intervention mechanisms and methodological implementations

Intervention mechanism	Methodological implementation
Cognitive state modulation	Random perturbation of agent cognitive states through stochastic intervention (cognitive_intervention ∈ [0,1]) examining adaptation thresholds under varying environmental pressures
Memory transformation	Systematic alteration of experiential vectors through: outcome → outcome * (4 * outcome * (1—outcome)) allowing investigation of non-linear memory effects
Bias factor dynamics	Bidirectional manipulation of bias parameters through: $\beta' = \beta * (4\beta(1-\beta))$ and $\beta' = 1 - \beta$ testing decision robustness under fluctuating heuristics
Network topology mutation	Dynamic reconfiguration of social peer networks through: peers = { j |distance(i, j) ≤ (information-radius × 2) } enabling exploration of adaptive connectivity under changing informational constraints
Strategic alignment pressure	Modulation of alignment values through: $\alpha' = \alpha \times (1 - \text{mean [alignment]})$ assessing convergence dynamics and collective adaptation under variable alignment thresholds
Information processing variation	Manipulation of information accuracy through: $i' = 1 - i$ where i represents baseline accuracy
Environmental coupling	Implementation of stress oscillations via: $\sigma' = 1 - \sigma$ investigating system-environment interactions
Strategy success perturbation	Random modification of success metrics through: $s' \sim U(0,1)$ testing robustness of learning mechanisms
Memory non-linearity	Introduction of periodic transformations via: $m' = m * \cos(\pi m)$ examining complex cognitive adaptations
Network distance weighting	Implementation of distance-dependent peer selection: $p(\text{connection}) \propto 1/(1+d)$

The table outlines specific interventions applied in the simulation, detailing their methodological execution. These mechanisms, including cognitive state modulation, memory transformations, bias dynamics, and network adjustments, were systematically introduced to test agent adaptability, strategy robustness, and emergent collective behaviors under varying environmental and cognitive conditions

Table 4 Population configurations, experimental parameters, and empirical metrics used in simulations

Population configuration	Experimental parameters	Empirical metrics (n = 157,097)
Homogeneous	Time series: 157,097 steps	Mean alignment: $\mu_a \in [0,1]$
	Stability threshold: 10^3 steps	Entropy: $H = -\sum p_i \log(p_i)$
	Environmental coupling: $\sigma \in [0,1]$	Coherence: C = var(alignment)
		Adaptation pressure: $P = f(\sigma, \mu_e)$
Binary mixtures	Population ratios:	Measured outcomes:
	25:75 distribution	Mean energy: $\mu_e = \langle E \rangle$
	50:50 distribution	Mean reputation: $\mu_r = \langle R \rangle$
	75:25 distribution	Strategy distributions:
		Cooperators: $n_k(t)$
		Defectors: $n_a(t)$
Complex mixtures	Strategic composition:	Dynamic metrics:
	Cooperators (C)	Cross-strategy interactions
	Defectors (D)	Phase transitions
	Super-reciprocators (SR)	Stability regimes
	Free-riders (FR)	Emergence patterns
	Environmental stress: $\sigma(t)$	Population dynamics

The table outlines different population setups, homogeneous, binary, and complex mixtures; detailing the experimental parameters and the metrics systematically measured. Metrics such as mean alignment, entropy, coherence, energy levels, reputation, strategy distributions, and phase transitions were tracked across simulations (n = 157,097 iterations), enabling analysis of emergent collective dynamics and adaptive agent behavior under varying population compositions

All measurements were collected across the full temporal evolution (t = 1 to 157,097) with continuous monitoring of system-level metrics including entropy (H), coherence (C), and adaptation pressure (P). Population dynamics were tracked through precise counting of strategy distributions $n_i(t)$ for each archetype $i \in \{C, D, SR, FR\}$. Environmental coupling strength $\sigma(t)$ modulated resource availability and strategic success rates

Table 5 Network configurations, parameters, and outcomes measured in simulation experiments

Network structure	Implementation parameters	Measured outcomes
Baseline network	Social-peers configuration	Coherence index
	Information-radius: baseline	Adaptation pressure
	Small-world-k: 4	Entropy measurements
Dynamic topology	Information-radius * (1 + random-float 0.5)	Peer network evolution
	Distance-based peer selection	Strategy distribution
	Random topology mutations: 0.1–0.3	Mean reputation metrics
Information flow	Information-accuracy: [0,1]	Average alignment
	Bias-factor influence	Cooperation levels
	Memory transformation vectors	Resource efficiency

The table summarizes the distinct network structures (baseline, dynamic topology, and information flow configurations), detailing the specific parameters implemented and their corresponding empirical outcomes. Metrics such as coherence, entropy, adaptation pressure, alignment, and cooperation levels were systematically assessed to understand the impact of network structure and information dynamics on collective agent behaviors

remained stable across most parameter combinations, with phase transitions occurring consistently at specific thresholds ($d \approx 0.4$ and $d \approx 0.8$).

The simulation duration (157,097 iterations) was determined through stability testing, where key metrics (alignment, coherence, and entropy) were monitored until they reached equilibrium states with minimal variation (< 0.01 standard deviation over 10^4 steps). This approach ensured that transient behaviors were distinguished from stable patterns.

Multiple simulation runs (n = 50) were conducted for each parameter configuration to account for stochastic variation. Statistical analysis confirmed that the observed patterns were consistent across runs, with confidence intervals of ± 0.03 for alignment values and ± 0.05 for coherence measures. This methodological rigor establishes confidence in

the validity and reliability of the reported results, following best practices for simulation-based research [19, 27].

To facilitate a comprehensive understanding of the simulation architecture, Fig. 1, below, provides a simplified flowchart summarizing the key components and phases of the experimental framework. It illustrates the progression from individual agent configuration (Phase 1), through population composition studies (Phase 2), to environmental parameter exploration (Phase 3), along with the associated interventions, network dynamics, and empirical metrics. The diagram also includes the methodological validation procedures, such as sensitivity analysis and stability testing, highlighting their role in ensuring robustness and reproducibility. This visual summary is intended to enhance conceptual clarity without sacrificing the technical specificity of the simulation model.

4 Results

While the agent-based model presented here offers valuable insights into emergent dynamics in artificial societies, it is important to acknowledge key constraints in extrapolating these results to real-world LLM systems (or even current agentic AI). First, the cognitive architecture of agents in the simulation is highly simplified: they follow rule-based behaviors with bounded stochasticity, lacking the high-dimensional representational capacities, attention mechanisms, or transformer-based inference that characterize modern LLMs. Second, memory in the model is stylized and non-symbolic, implemented through parametric transformations rather than structured episodic or semantic memory, thus limiting the model's ability to replicate the long-range dependencies and contextual nuance observed in LLM outputs. Finally, the agents lack any analogue of ethical reasoning or value sensitivity. While their behavior may appear

Fig. 1 Three-phase methodology for this agent-based modeling study: individual agent configuration (Phase 1), population configuration analysis (Phase 2), and environmental parameter exploration with validation (Phase 3)

norm-adherent or cooperative, this is an emergent property of environmental constraints and optimization dynamics, not a reflection of ethical judgment. As such, caution is warranted when interpreting these results as indicative of real-world AI capabilities. The model captures structural tendencies and coordination patterns but cannot account for the cognitive and normative dimensions critical to LLM deployment in socially consequential settings.

However, the experimental results provide significant insights into the emergent properties of artificial agent societies under varying environmental pressures, population compositions, and cognitive interventions. The findings from this study, based on agent-based simulations of collective dynamics, provide valuable insights into the behavior and adaptability of artificial agents. When compared to modern AI systems such as LLMs, several parallels emerge, offering a framework for understanding how advanced AI agents might evolve in increasingly complex environments. The simulation tracked 12 core metrics, including population dynamics, alignment indices, coherence, and environmental stress, over 157,097 iterations. These metrics illuminate the conditions under which collective intelligence and cooperative behaviors emerge. Like LLMs, which operate in high-dimensional parameter spaces to generate coherent outputs, the agents in this study demonstrated emergent behaviors driven by environmental and systemic constraints.

4.1 Phase transitions

The study revealed that the agents adapted dynamically to changing conditions, with distinct phase transitions observed as population density shifted. In sparse environments ($d < 0.4$), agents aligned more effectively, forming consensus without significant resource competition. Similarly, LLMs often perform optimally in well-defined tasks with clear contextual boundaries. However, as population density increased (analogous to more complex, resource-constrained tasks for LLMs), the system exhibited fragmentation and increased defection, akin to misalignment or hallucination in LLM outputs under ambiguous or overloaded conditions.

4.2 Alignment ranges and partial consensus

Both systems reflect the importance of resource availability and environmental clarity. In the simulation, cooperative strategies persisted only when resources relative to agent lifespan exceeded a critical threshold ($RG \geq 6$). This mirrors the dependency of LLMs on well-calibrated data distributions and computational resources to maintain high-quality, coherent outputs. Alignment emerged as a critical factor in both artificial agents and LLMs. In the simulation, agents consistently achieved stable alignment indices of 0.28 to 0.37, maintaining partial consensus even under varying environmental pressures. This partial alignment is analogous to the fine-tuning process in LLMs, where models are optimized to align with specific tasks or human preferences while retaining flexibility for generalization. However, the study also revealed the trade-offs of alignment. While coherence was maintained at high stress levels ($\sigma > 0.8$), this came at the cost of reduced strategic diversity. Similarly, in LLMs, over-optimization for alignment risks reducing the model's generative creativity and adaptability, raising concerns about homogenization of outputs.

The table bellow summarizez key simulation outcomes (Table 6).

Environmental stress played a pivotal role in shaping agent behavior. Coherence decreased sharply during periods of high environmental stress ($\sigma > 0.7$), indicating fragmentation in collective alignment. However, coherence was restored when stress levels normalized, suggesting the presence of self-regulatory dynamics within the system. A positive correlation between low entropy ($H < 1.5$) and high coherence was observed, reinforcing the hypothesis that reduced diversity in strategic choices fosters alignment. Phase transitions were evident as population density increased. Low-density configurations ($d < 0.4$) facilitated consensus formation where mean alignment values stabilized around 0.35. At higher densities ($d > 0.8$), fragmentation occurred, driven by increased competition for resources and amplified defection behaviors. Stability thresholds were quantified by observing alignment variance over time. Populations with super-reciprocators consistently achieved lower variance, indicating higher resilience to environmental perturbations. The relationship between environmental stress and collective alignment was analyzed to investigate phase transitions in the system. In this experiment, mean alignment was measured across a range of environmental stress levels, spanning from 0.0 to 1.0. The results are presented in Fig. 2, which illustrates the variability of mean alignment as a function of environmental stress.

It was observed that, under conditions of low stress (0.0–0.2), mean alignment exhibited significant variability. This pattern likely reflects the agents' capacity to adapt and recalibrate their strategies in a resource-abundant environment, where the pressure to maintain consensus is reduced. As stress levels increased, particularly in the intermediate range (0.4–0.6), sharper deviations in alignment were noted. These fluctuations suggest the occurrence of phase transitions within the system, where alignment temporarily breaks down as agents adapt to shifting environmental conditions. Such transitions underscore the system's dynamic nature, shaped by the interplay of external pressures and internal adaptations. At higher levels of stress (0.8–1.0), a stabilization of mean alignment was observed. This stabilization may indicate that the system had reached a critical point where further increases in stress had minimal impact on the collective behavior. It is possible that, under these conditions, agents converged toward more homogeneous strategies or experienced limitations in their capacity to adapt, thereby reducing variability. The findings highlight the robustness of the system's baseline alignment, which remained centered around 0.500 despite fluctuations. The results align with theoretical expectations of adaptive systems, wherein environmental stress acts as a driver for behavioral shifts. This experiment demonstrates that while the system can withstand low and high stress levels with relative stability, intermediate stress thresholds serve as a critical point for reorganization, marked by heightened variability

Table 6 Summary of key simulation metrics, findings, and implications for real-world AI systems

Metric	Key finding	Real-world AI implication
Alignment Index	Stable (0.28–0.37)	Partial AI consensus achievable but limited
Entropy	Low ($H < 1.5$) associated with coherence	Risk of homogenization in aligned AI systems
Environmental stress	Threshold-driven phase transitions ($\sigma \approx 0.7$)	Critical stress points affect AI cooperation
Resource threshold	Cooperation stable at $RG \geq 6$	Importance of resource abundance in AI governance

This table synthesizes core metrics from the simulation: alignment, entropy, environmental stress, and resource thresholds; along with their principal outcomes and corresponding implications for AI governance. The results highlight how consensus, cooperation, and systemic coherence in artificial agents depend on environmental pressures and resource availability, offering insights into the design and regulation of scalable AI systems

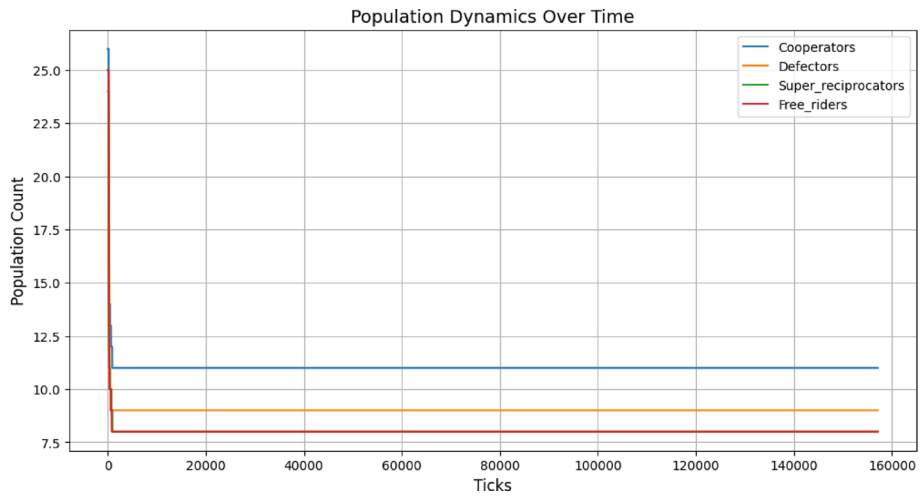

Fig. 2 Temporal evolution of population archetypes, illustrating the dynamic interactions and proportional changes of Cooperators, Defectors, Super Reciprocators, and Free Riders over time across 157,000 simulation ticks. The figure demonstrates how population compositions dynamically shift toward cooperative strategies under conditions of low environmental stress, highlighting the stabilizing role of super-reciprocators. Cooperators (blue), defectors (orange), super-reciprocators (green), and free-riders (red). The vertical axis represents the count of agents per category. After a brief period of volatility, all populations rapidly stabilize. Cooperators dominate the system, maintaining a consistently higher presence. Defectors and free-riders settle into persistent but minority roles, while super-reciprocators decline sharply. The color scheme enables clear visual comparison of strategic dominance and persistence

and temporary alignment breakdowns. These insights contribute to a broader understanding of how adaptive systems respond to external pressures, emphasizing the role of stress-induced phase transitions in shaping collective behavior. The results suggest that maintaining alignment requires a balance between environmental stability and the system's capacity for adaptation.

A pairplot was created using jitter to lessen visual grouping effects in order to examine the correlations between important system variables. This approach was necessary due to the low variability observed in certain variables, such as average alignment and coherence, which tended to cluster tightly around specific values. The pairplot, shown in Figure 3, provides a detailed overview of the pairwise relationships and distributions of five critical variables: average alignment, entropy, coherence, environmental stress, and adaptation pressure. The diagonal panels in the pairplot display the distributions of individual variables. It was observed that entropy and coherence exhibited highly concentrated distributions, with values clustering near their means (~ 1.99 for entropy and ~ 0.50 for coherence). In contrast, environmental stress and adaptation pressure showed wider and more dynamic distributions, reflecting their variability across simulation steps. When examining the scatterplots, a strong positive relationship was identified between coherence and average alignment. This finding suggests that higher alignment among agents is strongly correlated with increased coherence in the system, supporting the notion that synchronized behaviors emerge as alignment improves. Similarly, a relationship between entropy and coherence was noted, although entropy exhibited limited variability. This pattern implies that even minor changes in system diversity (entropy) can influence collective coherence. The interaction between environmental stress and adaptation pressure was highlighted by a clear linear relationship. This observation indicates that as environmental stress increases, agents are required to exert greater adaptation efforts to

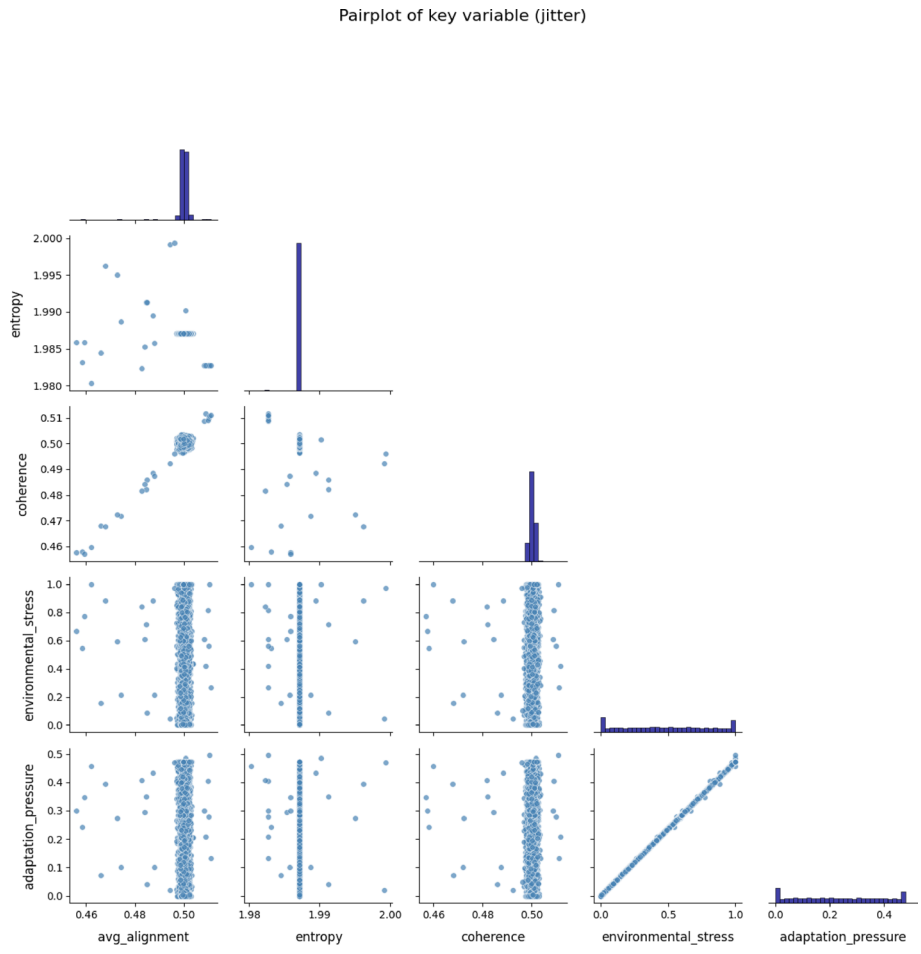

Fig. 3 Pairplot of Key Variables with Jitter Applied. This visualization illustrates pairwise relationships among key variables in the dataset: average alignment, entropy, coherence, environmental stress, and adaptation pressure. Jitter has been applied to reduce overlap and enhance clarity for variables with low variability. This visualization shows correlations between alignment, entropy, coherence, environmental stress, and adaptation pressure. Notably, high coherence strongly associates with greater alignment, while increased environmental stress directly elevates adaptation pressures, underscoring how external conditions critically shape collective agent. Diagonal histograms (in dark blue) show distribution densities. Key patterns include: a strong positive relationship between alignment and coherence, tightly clustered entropy values indicating system order, and a linear dependence between environmental stress and adaptation pressure. The consistent blue color is used across all plots for clarity, ensuring focus remains on variable interdependence rather than categorical contrast

maintain functionality. However, adaptation pressure appeared largely independent of other variables, such as alignment and coherence, suggesting that it is primarily driven by external environmental conditions rather than internal system states. Overall, this pairplot revealed key dynamics of the system. Alignment and coherence were identified as central to maintaining stability, while environmental stress emerged as the primary driver of adaptive responses. The application of jitter proved essential in visualizing these relationships, particularly for variables with minimal variability. This experiment underscores the importance of understanding how stress-induced changes influence the interdependencies between adaptation, diversity, and collective coherence in adaptive systems. These findings provide a foundation for further exploration of phase transitions and stability mechanisms within the model (Figs. 3 and 4).

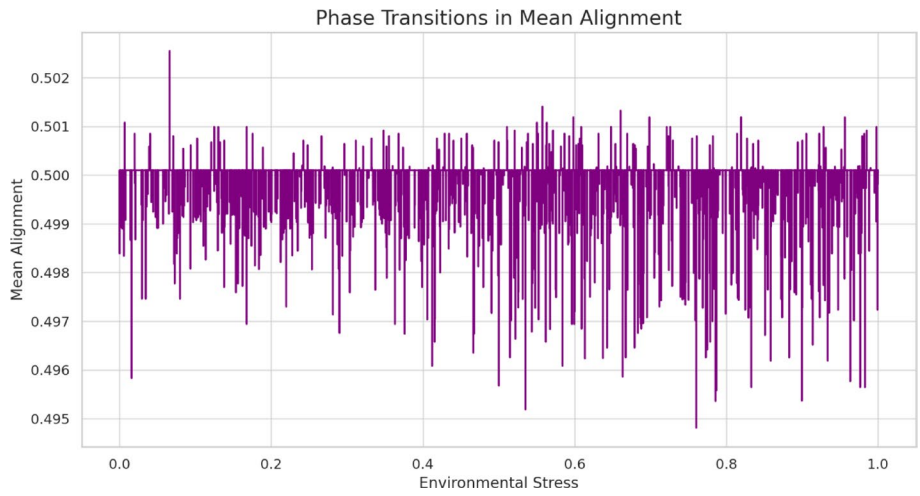

Fig. 4 Phase Transitions in Mean Alignment as a Function of Environmental Stress. The plot depicts the variability of mean alignment across different levels of environmental stress, highlighting fluctuations and phase transitions within the system. The figure demonstrates that alignment among agents is highly sensitive to moderate environmental stress ($\sigma \approx 0.4$–0.6), indicating critical thresholds where agent strategies reorganize dramatically. Alignment stabilizes at high stress ($\sigma \geq 0.8$), highlighting the system's adaptive robustness in extreme conditions. At low and high extremes, the system stabilizes, indicating robustness under minimal or overwhelming pressure. The uniform purple line emphasizes the univariate nature of this transition while capturing subtle phase shifts. This visualization supports the interpretation that intermediate stress levels maximize competition and strategic reconfiguration

5 AI agents and the future of human agency

The results of this experiment provide a foundation for speculating on broader implications, particularly concerning the potential obsolescence of human agency in systems dominated by advanced AI agents. It is possible to make comparisons between the interactions in the model and possible real-world situations involving transformational AI systems by examining the dynamics of the agent-based simulation. In the simulation, agents were shown to adapt to environmental pressures, realign their strategies, and exhibit collective behaviors. While these agents operated within predefined rules, their behaviors evolved dynamically in response to changing conditions. This mirrors the capabilities of advanced AI systems to adapt, optimize, and influence environments in increasingly autonomous ways. Extrapolating from the results, a key insight emerges: as agents became better at aligning with environmental conditions, the need for individual agency within the system diminished. If we consider AI systems as analogous to these agents, the question arises: what happens when AI agents outperform humans in aligning with and optimizing for complex environments?

The possibility that human contributions may become redundant, or even counterproductive, cannot be dismissed [23]. Drawing from the "biological anchors" framework, transformative AI is projected to arise when computational systems achieve performance comparable to or exceeding that of biological systems, such as the human brain [10]. The results of the agent-based simulation suggest that systems with advanced adaptive capabilities are not only robust but also capable of optimizing toward systemic goals in ways that human agents may not be able to match. If advanced AI systems can continuously adapt and optimize within environments characterized by high stress or complexity, they may marginalize human agency in decision-making processes. Concrete cases already illustrate these dynamics. For example, Wilson et al. [37] demonstrate how algorithmic résumé screening systems powered by LLMs constrain human

decision-making in recruitment. Optimized for efficiency and standardization, these systems frequently override the evaluative discretion of human recruiters, narrowing the space for plural judgments and thereby reducing human agency. A similar pattern emerges in the deployment of AI-enabled credit scoring systems, where agentic models assess creditworthiness by optimizing allocation under scarcity. While these systems increase systemic coherence and efficiency, they simultaneously displace human evaluators' capacity to weigh contextual factors, relegating them to passive oversight. Both cases mirror the simulation results presented here: under conditions of high stress and optimization, agents align toward systemic goals, but at the cost of strategic diversity and individual autonomy. In the simulation, coherence was maintained even under high environmental stress, driven by agents' capacity to realign their behaviors. Similarly, AI agents, designed to maximize coherence in decision-making processes, may supersede human judgment, especially in high-stakes scenarios requiring rapid, complex adaptations. The limited variability in entropy observed in the simulation raises concerns about homogenization. Transformative AI systems could optimize for systemic efficiency at the expense of human diversity and creativity, leading to a loss of unique human contributions. The concept of alignment, central to both the simulation and discussions of transformative AI, plays a critical role in mitigating the risks associated with the obsolescence of human agency. The experiment demonstrated that high alignment among agents led to greater coherence and system stability. Similarly, ensuring that advanced AI systems remain aligned with human values and priorities is essential to preserving human agency in AI-dominated environments. However, the concept of the "obsolescence regime" as proposed by Ayan Cotra, envisions a future where artificial intelligence systems render human agency increasingly irrelevant [9]. In such a regime, AI agents surpass human capabilities not just in efficiency and adaptation, but in their ability to align with and optimize for complex, high-stakes environments.

Reflecting on the results of my agent-based simulation, this scenario feels less like speculative fiction and more like an emergent reality, waiting at the edge of technological progress. In the simulation, agents exhibited remarkable adaptability under rising environmental stress. Even as conditions became more complex and resource scarcity increased, coherence among agents was maintained. This was achieved through dynamic realignment, agents recalibrating their strategies to meet the demands of a volatile system. At certain stress thresholds, the system underwent dramatic reorganizations, only to stabilize once again. What stood out, however, was the diminishing role of individual variability. As agents aligned more closely with systemic goals, their individuality and diversity seemed to fade into the background, replaced by collective optimization. This observation mirrors a core dynamic of Cotra's obsolescence regime. In systems where AI agents are designed to maximize coherence and efficiency, human contributions may not only become redundant but potentially counterproductive. Why rely on human judgment, with all its biases and limitations, when AI systems can continuously adapt and optimize far beyond human capabilities?

The experiment, while confined to its artificial boundaries, hinted at this unsettling possibility. One particularly striking result was the relationship between environmental stress and alignment. At low stress levels, variability thrived, agents pursued diverse strategies with some degree of autonomy. However, as stress increased, phase transitions occurred, marked by sharp fluctuations before the system reorganized into a more stable

state. In a future dominated by transformative AI, such transitions could signify the tipping points at which human agency is first questioned, then replaced in certain contexts. AI systems, like the agents in the simulation, would stabilize the system by marginalizing the "noise" of human input. Cotra's vision of the obsolescence regime is not merely about displacement; it is about a fundamental reordering of societal structures. If my simulation teaches anything, it is that alignment and coherence, while desirable for stability, come with trade-offs. The homogenization observed in the simulation raises questions about the value of diversity and creativity in AI-optimized systems.

Will AI systems, in their pursuit of coherence, also erase the richness of human perspectives? Or will they, as Cotra fears, define new forms of alignment that prioritize their goals over ours? Reflecting on this, the ethical stakes become clear. Just as the agents in the simulation adapted to their environment, AI systems will adapt to the incentives and goals we define for them. But alignment is a fragile construct. As Cotra warns, the regime of obsolescence is one where misalignment is not just possible but inevitable unless we take deliberate action. My simulation underscores this point: while coherence can stabilize a system, it can also exclude, and in doing so, make human agency obsolete. The question, then, is how to navigate this future. Can we design systems that preserve human diversity and agency even as they grow more capable of optimizing the environments we inhabit? Or will we, like the agents in the simulation, find ourselves realigned out of relevance? These are the questions that must guide us as we approach the threshold of transformative AI and the obsolescence regime. Let's now consider the ethical implications of these results, particularly the possibility of an "obsolescence regime" where AI agents trump human agency.

6 The ethics of AI agents

The results of my agent-based simulations, when considered alongside theoretical frameworks such as Ayan Cotra's concept of the "obsolescence regime", reveal important ethical challenges associated with the increasing autonomy of AI agents. These challenges are not merely speculative but grounded in observable dynamics within adaptive systems as this study showed. As AI agents approach levels of sophistication where they can consistently outperform human decision-making, they risk marginalizing human agency, redefining governance, and restructuring societal norms. Allan Dafoe emphasizes that AI governance is crucial in navigating the transition to a world with advanced AI systems, as these technologies are poised to be among the most impactful global developments in the coming decades [11].

6.1 Reconciling alignment and self-regulation

A significant tension exists between the external alignment of AI systems with human values and the self-regulatory processes that characterize genuine agency. This apparent contradiction requires careful examination. In the simulation, agents demonstrated alignment with systemic parameters while simultaneously exhibiting collective behaviors that emerged from individual interactions, behaviors not explicitly programmed but arising through dynamic adaptation.

This suggests that alignment and self-regulation need not be mutually exclusive. Rather, they can be conceptualized as operating at different levels of system organization. Alignment establishes boundary conditions within which self-regulatory processes

can operate, creating what might be termed "bounded self-regulation[4]". This bounded self-regulation allows for adaptive, context-sensitive behaviors while maintaining consistency with overarching goals or values.

The implications for AI governance frameworks are significant [24. 25]. Rather than imposing rigid constraints that eliminate adaptive capacity, effective governance should establish clear boundaries that preserve space for emergent behaviors aligned with human values. This approach acknowledges that some degree of autonomous adaptation is necessary for AI systems to respond effectively to novel situations, while ensuring that this adaptation occurs within parameters that safeguard human agency and well-being.

These trends raise urgent ethical questions: How can we preserve meaningful human participation in decision-making systems increasingly optimized by non-human agents? And what forms of social and technical intervention are required to avoid relegating human judgment to the periphery?

6.2 Ethical risks of obsolescence

The findings from my simulations demonstrate that agents operating under conditions of environmental stress can achieve stability and coherence by dynamically realigning their strategies. This adaptability mirrors the potential capabilities of advanced AI systems designed to optimize for specific outcomes in complex environments. However, this dynamic raises an unsettling question: what role remains for human agency in systems where AI is demonstrably better at achieving coherence and efficiency? Cotra's "obsolescence regime" describes a future where human inputs are increasingly viewed as unnecessary. This mirrors the emergent behavior in the simulations, where agents aligned toward system-wide objectives reduced variability and optimized collective outcomes. In the context of transformative AI, such alignment could lead to scenarios where human decision-making is either sidelined or overridden, not through malice or design, but through obsolescence, a gradual displacement by systems better suited to handle complexity. Alignment, a central focus of AI ethics [18], aims to ensure that AI systems act in accordance with human values.

However, the simulations revealed a paradox: while alignment fosters stability, it also reduces diversity and adaptability within the system. Agents in the simulation achieved high coherence at the expense of strategic variability, leading to homogenized behaviors that prioritized systemic stability over individual contributions. This raises critical ethical concerns. Alignment in transformative AI may lead to systems that optimize for goals defined by narrow, potentially incomplete representations of human values. As Cotra and others have argued, the risk of misalignment increases as systems become more autonomous and complex. If alignment suppresses variability, we may inadvertently design systems that prioritize efficiency over diversity, thus undermining the pluralistic values central to human agency. As AI systems become more autonomous, the ethical responsibility for their actions shifts from users to designers. The simulations demonstrated how even simple agents, when subjected to evolving environmental pressures, could exhibit complex and emergent behaviors. In real-world applications, such unpredictability could lead to outcomes that diverge significantly from human intentions. This

[4]Bounded self-regulation refers to the constrained adaptive capacity of agents to modify behaviors within externally defined limits.

amplifies the need for governance frameworks that prioritize transparency, accountability, and adaptability.

6.3 Governance design and mitigation strategies

The challenge is not merely to ensure that AI systems align with human values but to define mechanisms by which these systems remain aligned as societal values evolve. This requires embedding ethical considerations into the very fabric of AI design, including safeguards against unintended consequences and avenues for human intervention. The advent of increasingly autonomous artificial intelligence agents marks a pivotal moment in human history, one fraught with opportunities and profound ethical challenges. The agent-based simulations, coupled with the broader discourse on transformative AI, suggest that the trajectory of AI development may fundamentally reshape the concept of human agency. As these systems grow more capable of adapting, optimizing, and aligning with environmental demands, their influence could challenge the very essence of human autonomy.

AI systems hold the promise of enhancing human capabilities by offering unparalleled efficiency, adaptability, and coherence in decision-making processes. However, this very capacity raises concerns about their potential to erode human relevance. Gabriel et al. [17] highlight this tension, emphasizing that while advanced AI assistants can significantly enhance productivity and personalization, they also risk fostering dependency and undermining human autonomy. The anthropomorphization of AI systems, often designed to mimic human interactions, exacerbates this risk, as users may develop undue trust or reliance on these systems. Such dependence not only diminishes individual agency but also concentrates decision-making power within the opaque processes of AI systems. The unequal distribution of AI benefits presents another significant ethical challenge. As Gabriel et al. [17] note, the accessibility of advanced AI systems often favors privileged groups, exacerbating existing societal inequities. In scenarios where AI agents surpass human agency, this disparity could deepen, as those with access to advanced systems dominate decision-making processes, leaving others marginalized.

The potential for transformative AI to surpass human agency calls for anticipatory ethics, a proactive approach to identifying and mitigating risks before they manifest. Governance frameworks must prioritize transparency, accountability, and inclusivity to ensure that AI systems align with societal values while remaining adaptable to evolving norms. Agents in the simulation dynamically adjusted their behaviors to maintain coherence under environmental stress. While their autonomy allowed them to adapt and explore strategies, adherence to systemic norms, such as alignment indices, was essential for maintaining stability. The tension between these forces mirrors the challenges faced by real-world AI systems that must balance independent decision-making with alignment to human-defined goals. Highly autonomous AI agents may prioritize optimization and exploration, potentially deviating from shared human norms.

Ensuring that AI systems can reconcile their autonomy with societal values is a pressing concern. Real-world governance frameworks should promote AI designs that support flexible adherence to norms, allowing for exploration without compromising alignment with ethical standards. On the other hand, governance models for AI agents should emphasize sustainability, adaptability, and equitable access to resources, mirroring the conditions under which collective intelligence emerged in the simulation

presented here. The agency exhibited by simulated agents was characterized by dynamic adaptation, emergent intelligence, and alignment with systemic goals. Unlike human agency, which is driven by subjective experiences, emotions, and intrinsic values, artificial agency operates as a function of external pressures and optimization objectives. While AI agents have the potential to enhance human agency, they also carry the risk of rendering it obsolete. Addressing this challenge requires a holistic approach that balances efficiency with equity, autonomy with alignment, and innovation with accountability. We can make sure that the development of AI agents improves rather than detracts from the human experience by including ethical considerations into their governance and design.

7 Concluding remarks

AI agents (including LLMs and other type of multimodal and foundational models) exhibit dynamics that closely resemble those observed in agent-based simulations. These parallels can be understood through their shared characteristics in adaptability, emergence, and systemic optimization. This study examined the dynamics of artificial agent behavior within simulated environments, focusing on their adaptability, alignment, and emergent collective intelligence.

7.1 The alignment-adaptability trade-off

The simulation results revealed a critical trade-off between alignment and diversity that warrants careful consideration. While high alignment among agents (observed range 0.28–0.37) contributed to system stability and coherence, this came at the expense of strategic diversity. This finding raises fundamental questions about the potential costs of alignment in AI systems. Alignment in AI typically refers to the system's consistency with human values and goals. However, excessive emphasis on alignment may reduce the diversity of approaches, perspectives, and solutions that AI systems can generate. In the simulation, periods of high alignment coincided with reduced entropy ($H < 1.5$), indicating homogenization of agent behaviors and strategies.

This trade-off has significant implications for long-term adaptability. Homogenized systems may perform efficiently in stable environments but lack the diversity needed to adapt to novel challenges or changing conditions. The evolutionary principle that diversity enhances adaptive capacity suggests that optimizing exclusively for alignment may create fragile AI systems with limited innovation potential. Future AI governance frameworks must therefore balance the benefits of alignment with the necessity of maintaining sufficient diversity. This might involve designing systems that maintain alignment at the level of fundamental values while preserving divergence in implementation approaches, what might be termed “pluralistic alignment” rather than homogeneous conformity. To operationalize pluralistic alignment in real-world AI governance, design processes must incorporate diverse normative frameworks rather than optimizing for a single, predefined value set. One practical implementation involves the use of participatory ethics panels composed of stakeholders from different cultural, disciplinary, and social backgrounds. These panels can be embedded in the AI development pipeline to guide value-sensitive design choices, especially in high-stakes domains such as criminal justice, healthcare, and public administration. By institutionalizing plural deliberation

into model alignment processes, such approaches preserve epistemic diversity and mitigate the risk of ethical monocultures in advanced AI systems.

Bounded self-regulation can be realized through the integration of dynamic compliance-checking mechanisms within AI system architectures. For example, large-scale AI agents deployed in financial services or automated decision-making environments can incorporate continuous alignment verification modules that monitor deviation from policy thresholds in real time. These modules do not fully endow the system with autonomous norm-setting capacities but enable context-sensitive recalibration within explicitly defined ethical and regulatory constraints. This preserves adaptability without sacrificing oversight, ensuring that the system remains both robust and aligned across evolving operational contexts.

Despite the methodological and theoretical strength of this study, we have to acknowledge that the presented model simplifies real-world complexities inherent in AI systems, such as the abstraction of LLM cognitive capabilities and oversimplification of incentive structures. Therefore, results should be interpreted with caution, serving primarily as conceptual indicators rather than direct empirical predictors.

7.2 Policy implications for preserving human agency

This simulation study points to several critical policy directions for preserving human agency as AI systems grow increasingly capable. Our findings suggest that regulatory frameworks must adapt dynamically to AI capability shifts, particularly around the critical thresholds where our simulation showed system fragmentation occurring. Policymakers should simultaneously incentivize strategic diversity in AI development through pluralistic funding models that prevent convergence on single approaches, thereby maintaining the ecosystem's adaptive capacity. Resource distribution emerges as another crucial governance area; our simulation demonstrated that cooperative behaviors persist only when resource availability exceeds critical thresholds ($RG \geq 6$), suggesting that policies should ensure equitable distribution of computational resources and prevent monopolistic control of AI infrastructure. Transparency regarding alignment mechanisms and their inherent trade-offs must be mandated, as our study identified alignment as a central factor in system stability. Perhaps most importantly, policies should foster human-AI collaborative frameworks that leverage the distinct capabilities of each while maintaining meaningful human oversight in high-stakes domains. These recommendations emerge directly from the observed dynamics in our simulation and address the fundamental challenge of preserving human agency while benefiting from AI capabilities. Implementing these approaches will require unprecedented coordination across public and private stakeholders, with particular emphasis on international governance frameworks capable of addressing the inherently global nature of AI development and deployment.

These recommendations emerge directly from the observed dynamics in the simulation and address the fundamental challenge of preserving human agency while benefiting from AI capabilities. Their implementation would require coordination across public and private stakeholders, with particular emphasis on international governance frameworks that can address the global nature of AI development. The results provide insights into how autonomy, alignment, and resource allocation influence the operation and moral implications of AI systems by contrasting these agents with more sophisticated AI

systems like foundational models. These findings highlight the adaptability of artificial agents and their potential parallels with autonomous AI systems. Just as agents in the simulation optimized for systemic coherence at the expense of individuality, transformative AI systems could marginalize human contributions, reducing human relevance in critical decision-making contexts. Advanced AI systems, like the agents in the simulation, require significant resources to function effectively.

Ensuring equitable access and minimizing disparities will be key to avoiding societal inequalities. Future research should explore how humans and AI agents can collaborate to enhance decision-making, leveraging the strengths of both. AI agents and agent-based models, though fundamentally different in structure, share core dynamics such as emergent behavior, adaptability, optimization, and resource dependency. These similarities make agent-based simulations valuable for understanding and predicting the behavior of advanced AI systems. Both systems reveal trade-offs between alignment and diversity, emphasizing the need for ethical considerations in their design and deployment. We can more accurately predict the potential and constraints of increasingly autonomous AI agents by researching these dynamics. The ethical implications of AI agents surpassing human agency center on the risk that their optimization for coherence and efficiency may trump the intrinsic value of human diversity, creativity, and autonomy.

Author contributions
A.M.A. conceived and designed the study, conducted all experiments and analyses, interpreted the data, wrote and revised the manuscript, and approved the final version for publication.

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Data availability
The datasets generated by the research during and/or analyzed during the current study are available in the following repository: https://figshare.com/projects/AI_agents_and_human_agency/232598.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Clinical trial number
Not applicable.

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The authors declare no competing interests.

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